

Madhav Makwana

Software Engineer | Backend Systems · GenAI Applications · Machine Learning | DTU 2026

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Education

Delhi Technological University

2022 – 2026

B.Tech in Engineering Physics (Electronics), Minor in Computer Science — **CGPA: 7.14**

Experience

Appsoleut Games

June 2025 – Aug 2025

Data Science Intern

Gurugram

- Analyzed **10K+ gameplay sessions** to map early-player behavior patterns (retries, collisions, session duration) and identify high-churn business segments for product intervention.
- Trained a logistic regression classifier on engineered gameplay features, achieving **AUC > 0.80** on hold-out data.
- Delivered a **3-tier risk segmentation model** (low/medium/high) to guide onboarding strategy and retention decisions.

Projects

ResearchMind — Multi-Agent AI Research System

[🔗 Live Demo](#) | [🐙 GitHub](#)

Python, FastAPI, LangChain, Groq API, Tavily API, BS4, JWT, Prometheus, Docker, Streamlit

- Architected a **4-step LLM orchestration pipeline** integrating external web APIs (Tavily, BeautifulSoup) with an LLM backend for grounded, real-time report generation.
- Engineered a production-grade FastAPI backend with JWT auth, modular APIs, and JSON logging with trace IDs.
- Designed a Critic Chain self-evaluation loop scoring reports on accuracy and depth, reducing hallucination risk.
- Monitored pipeline with **Prometheus metrics** (per-step histograms, error counters), async test suite, deployed on Render.

YTLens — RAG-Powered YouTube Q&A System

[🔗 Live Demo](#) | [🐙 GitHub](#)

FastAPI, LangChain, FAISS, HuggingFace, JWT, SQLAlchemy, Docker, Streamlit

- Built an end-to-end **RAG pipeline** with multi-source transcript ingestion, structured chunking, **BGE embeddings**, and FAISS vector store persistence — effectively an ETL flow for unstructured media data.
- Replaced similarity search with **MMR retrieval** (k=5, fetch_k=20) to diversify context and improve answer grounding.
- Implemented a **3-layer transcript fetching strategy** (direct → proxy → manual fallback) ensuring zero hard failures.
- Hardened backend with bcrypt-hashed JWT auth, SQLAlchemy persistence, and shipped Dockerized service on Render.

Car Price Prediction API — Production-Grade ML System

[🔗 Live Demo](#) | [🐙 GitHub](#)

Python, FastAPI, scikit-learn, Redis, Docker, Prometheus, Grafana

- Developed a **production REST API** with FastAPI exposing real-time ML inference via a joblib pipeline, designed for seamless integration with downstream client systems.
- Enforced dual-layer auth (JWT + API key validation) and custom logging middleware for request tracking.
- Integrated **Redis caching** for repeated predictions, reducing redundant ML inference calls and improving latency.
- Set up **Prometheus + Grafana** observability (request counts, latency histograms); load-tested with **Locust** and shipped on Render via Docker.

Skills

Languages: Python, C++, SQL

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems

Backend: FastAPI, REST APIs, AsyncIO, Pydantic, SQLAlchemy, JWT Auth, OAuth2, Middleware, Logging

Web & Integration: HTML, CSS, API Integration, Data Pipelines, Workflow Automation

Databases & Caching: MySQL, Redis, SQLite

DevOps & Infra: Docker, Prometheus, Grafana, Git, GitHub, Render, Locust

ML & GenAI: scikit-learn, TensorFlow, Keras, LangChain, FAISS, ChromaDB, HuggingFace, OpenAI API, Groq API, RAG Pipelines, Agentic AI

Tools: Streamlit, Postman, Linux

Achievements

- Ranked among **top 5%** in NPTEL Introduction to Database (Q4 toppers list) [\[CERTIFICATE\]](#) [🔗](#)
- Awarded **Proficiency in Computer Science** in Class 12 [\[CERTIFICATE\]](#) [🔗](#)
- SOF Olympiads (top ~5% globally, qualified Level 2): **IEO Delhi Rank 19** [\[CERT\]](#) [🔗](#); **IMO Int. Rank 4033** [\[CERT\]](#) [🔗](#); **NSO Int. Rank 5065** [\[CERT\]](#) [🔗](#)